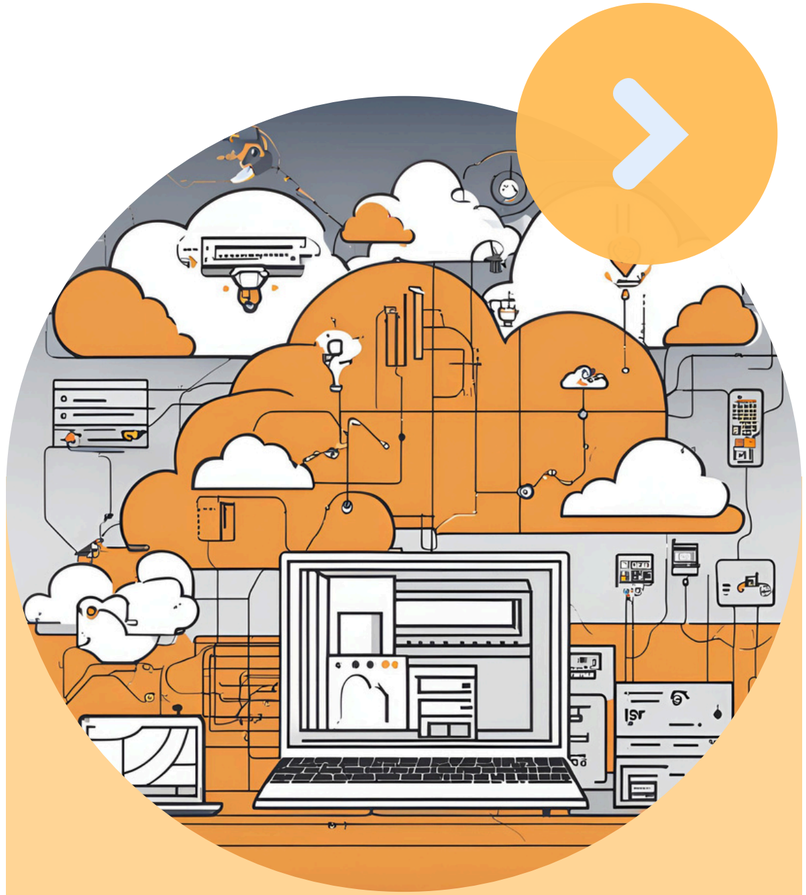
PLURALCODE ACADEMY

# STATIC WEBSITE HOSTING WITH AWS (GREENFIELD AGENCY)

## CLOUD COMPUTING - AUGUST 2024 COHORT

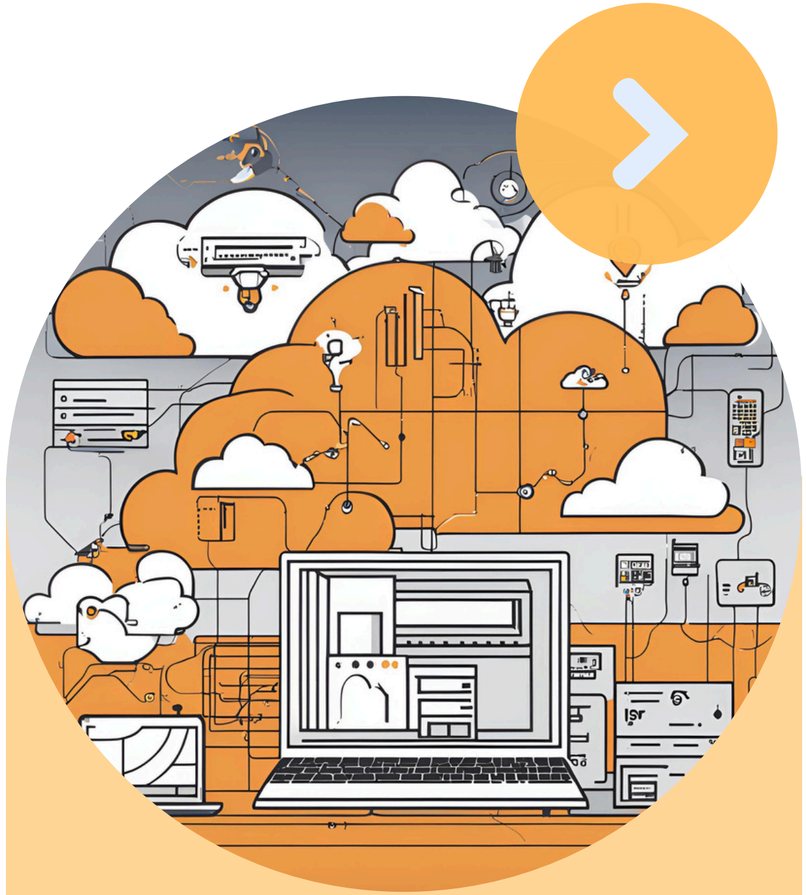
TOLUWANIMI AKINYEMI





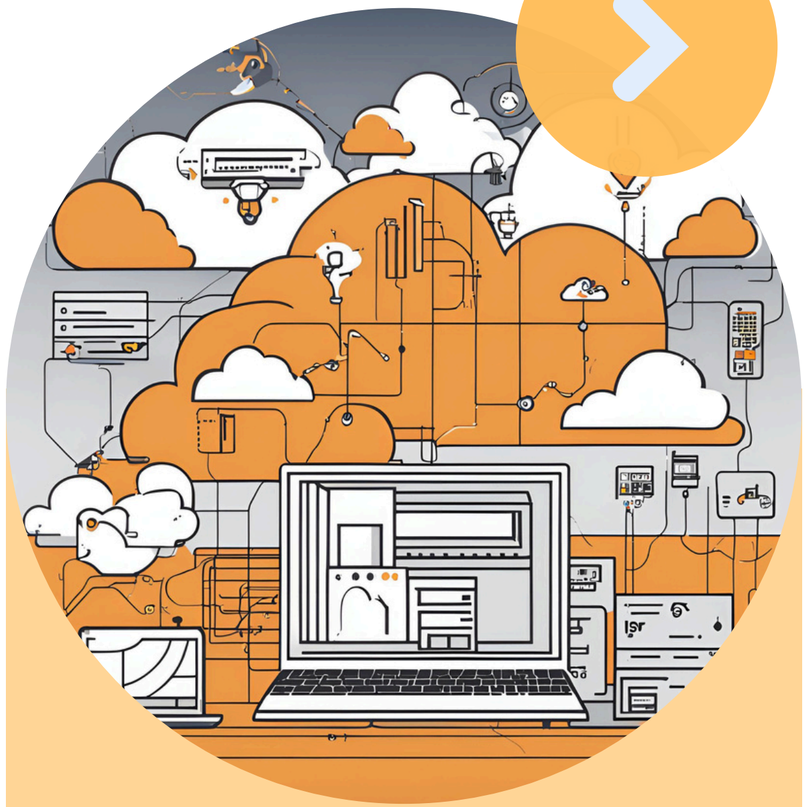
# Introduction

GreenField Agency, a digital marketing firm, recently decided to showcase its work and portfolio on a website. They were looking for a cost-effective way to host a static website with fast global delivery. After evaluating several options, they decided to host their website on AWS



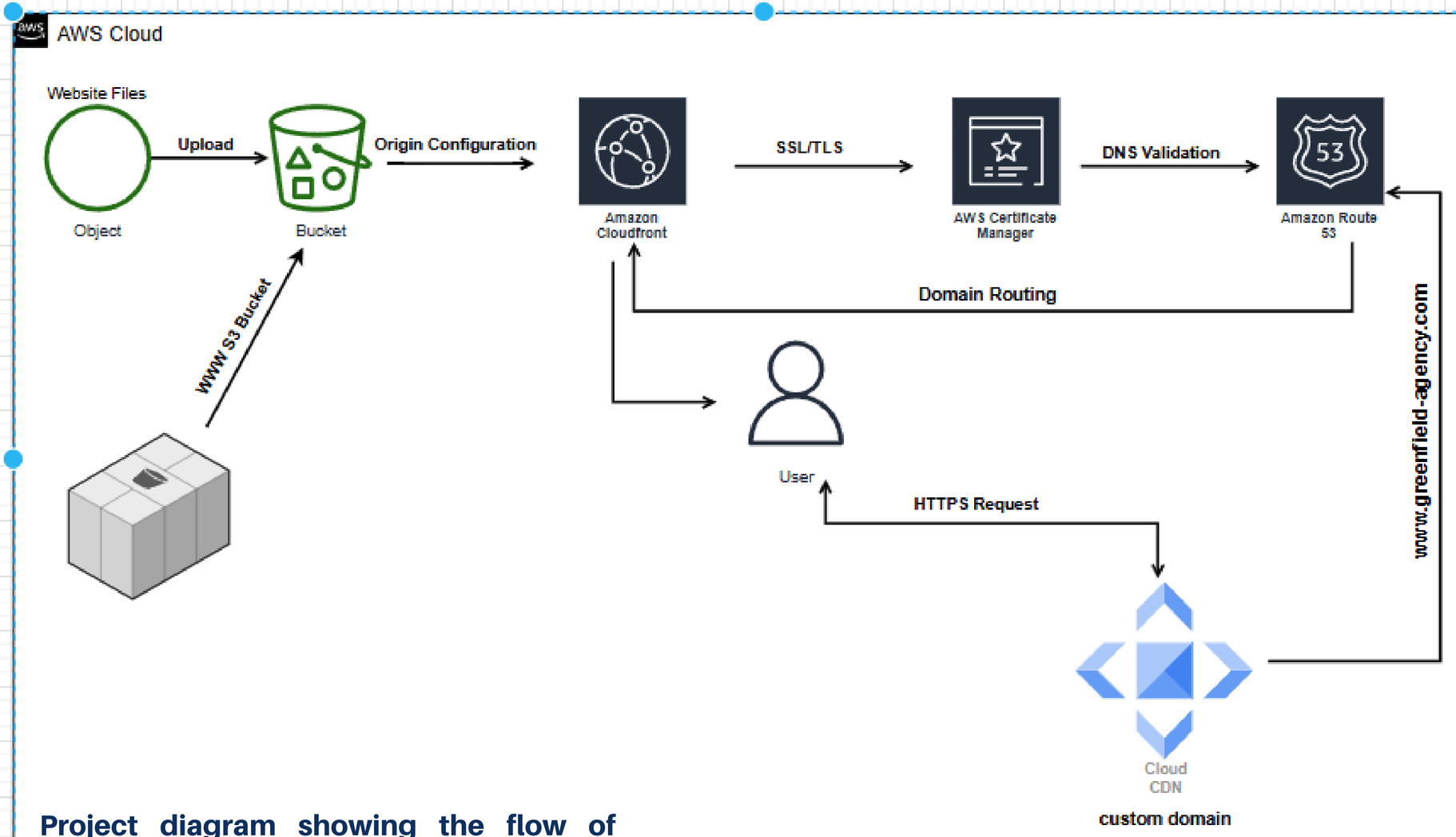
# Objective

The goal is to create a static website that can scale automatically based on demand, with a strong emphasis on speed and security. They plan to integrate SSL for secure browsing and configure a custom domain for their professional brand.

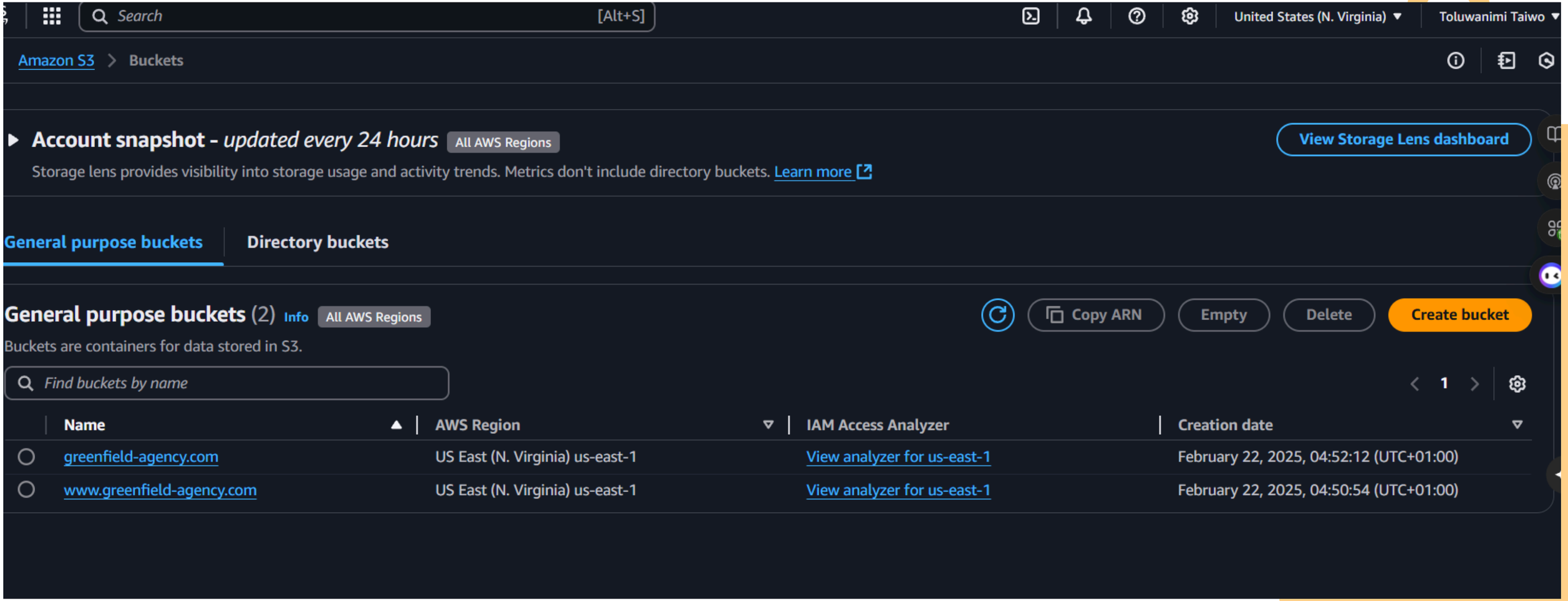


# Service Used

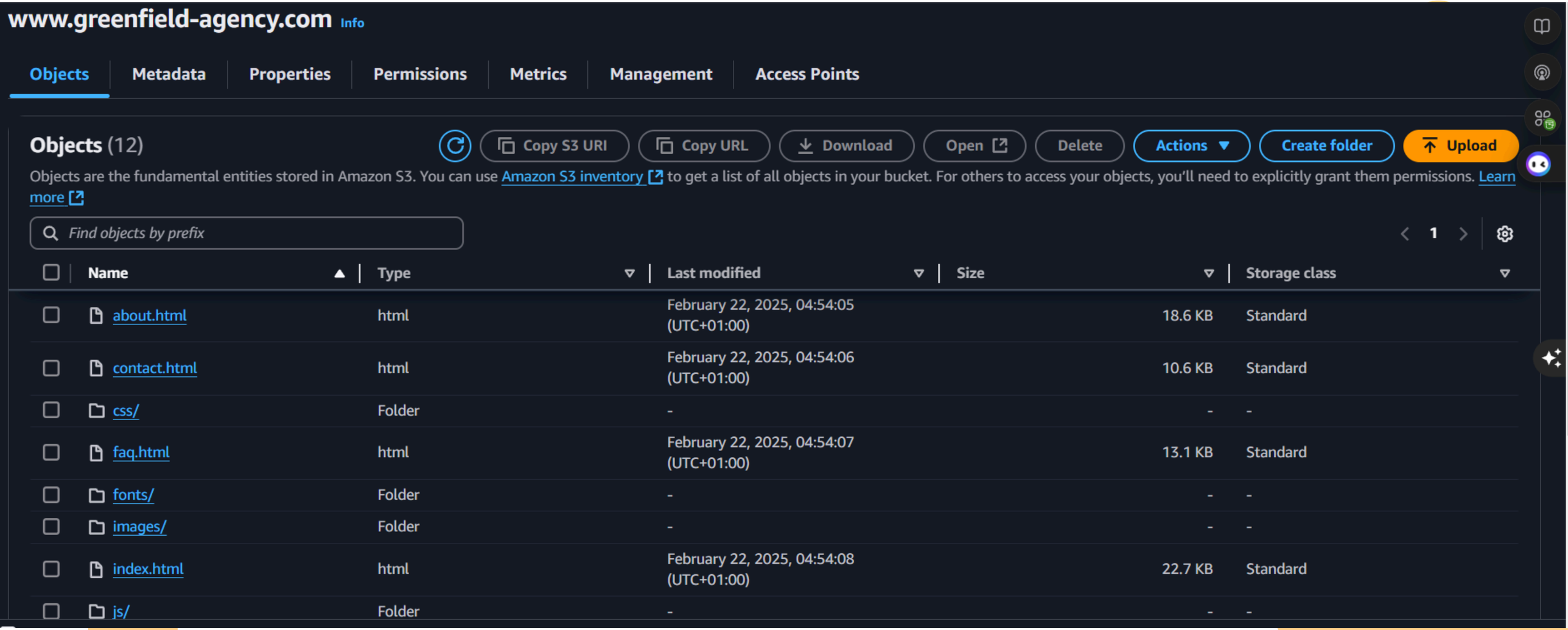
- Amazon S3: Hosts the static website content like HTML, CSS, and JavaScript files.
- Amazon CloudFront: Distributes the website globally, ensuring low-latency access for users across different regions.
- Amazon Route 53: Manages the custom domain and handles DNS routing for the website.
- AWS Certificate Manager: Provides SSL/TLS certificates to secure the site’s traffic.



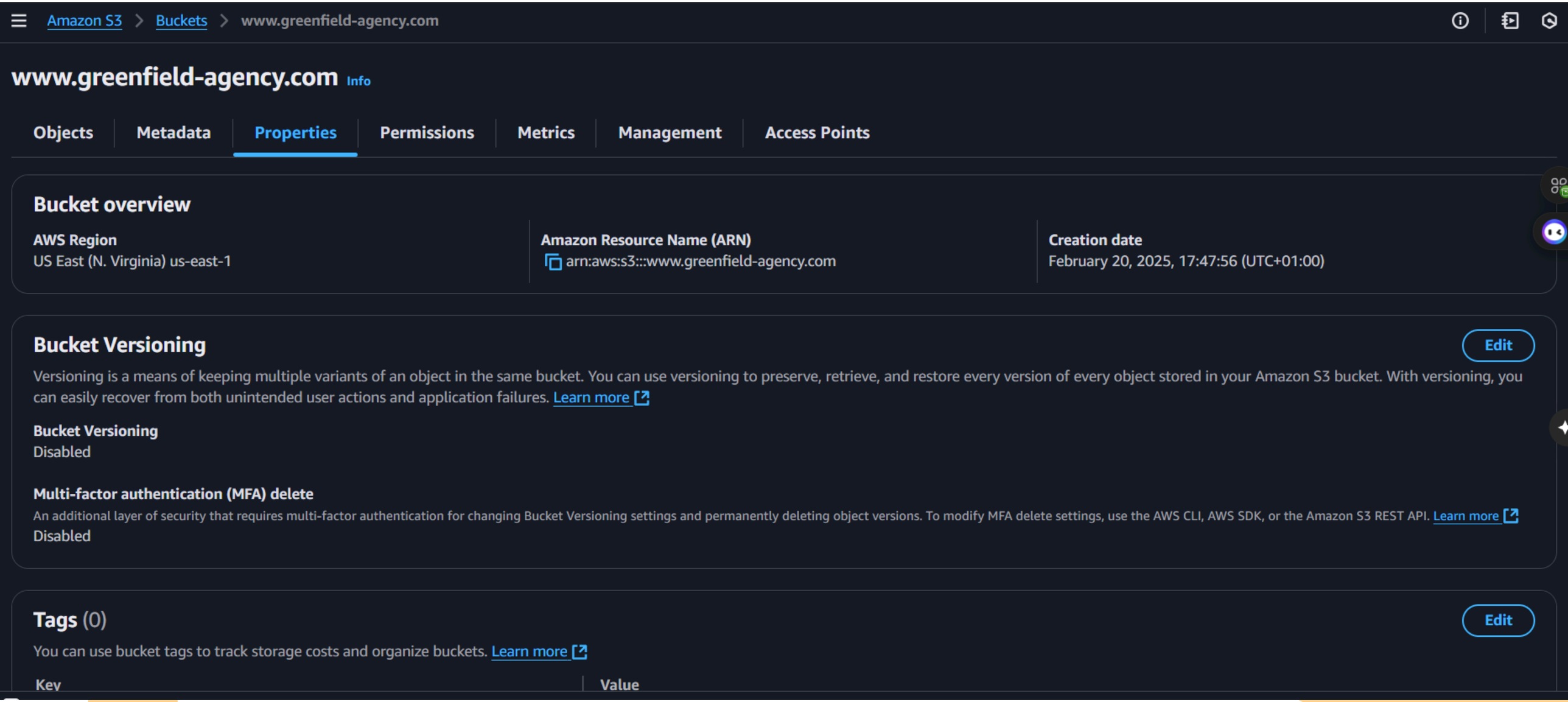

**Project diagram showing the flow of process of the project**



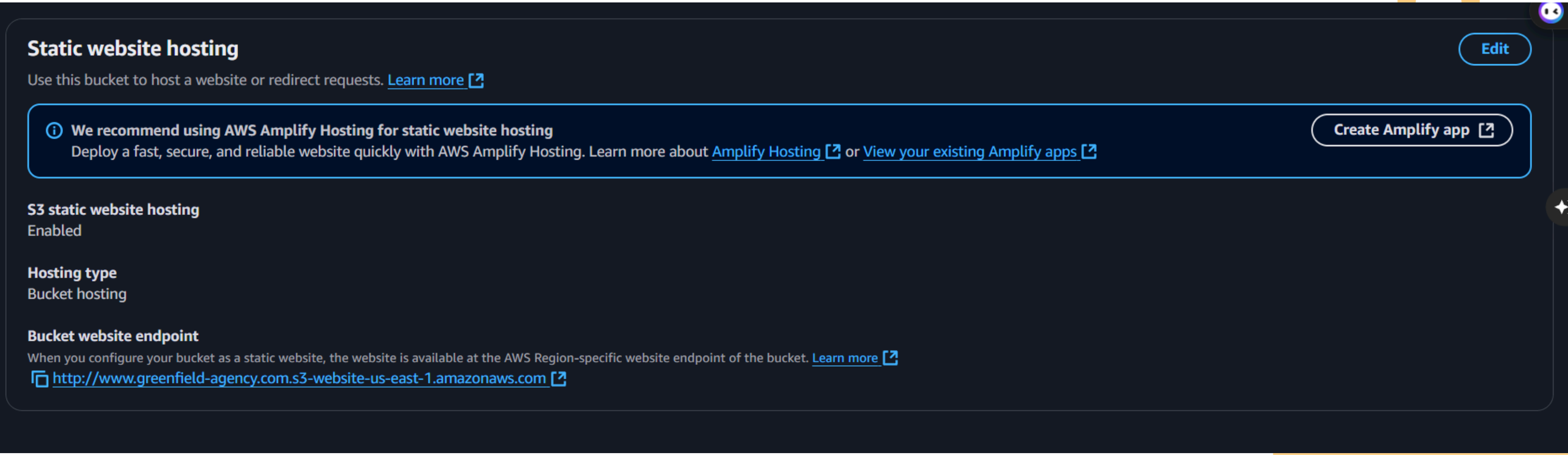

creation of S3 buckets




uploading the website files and folders



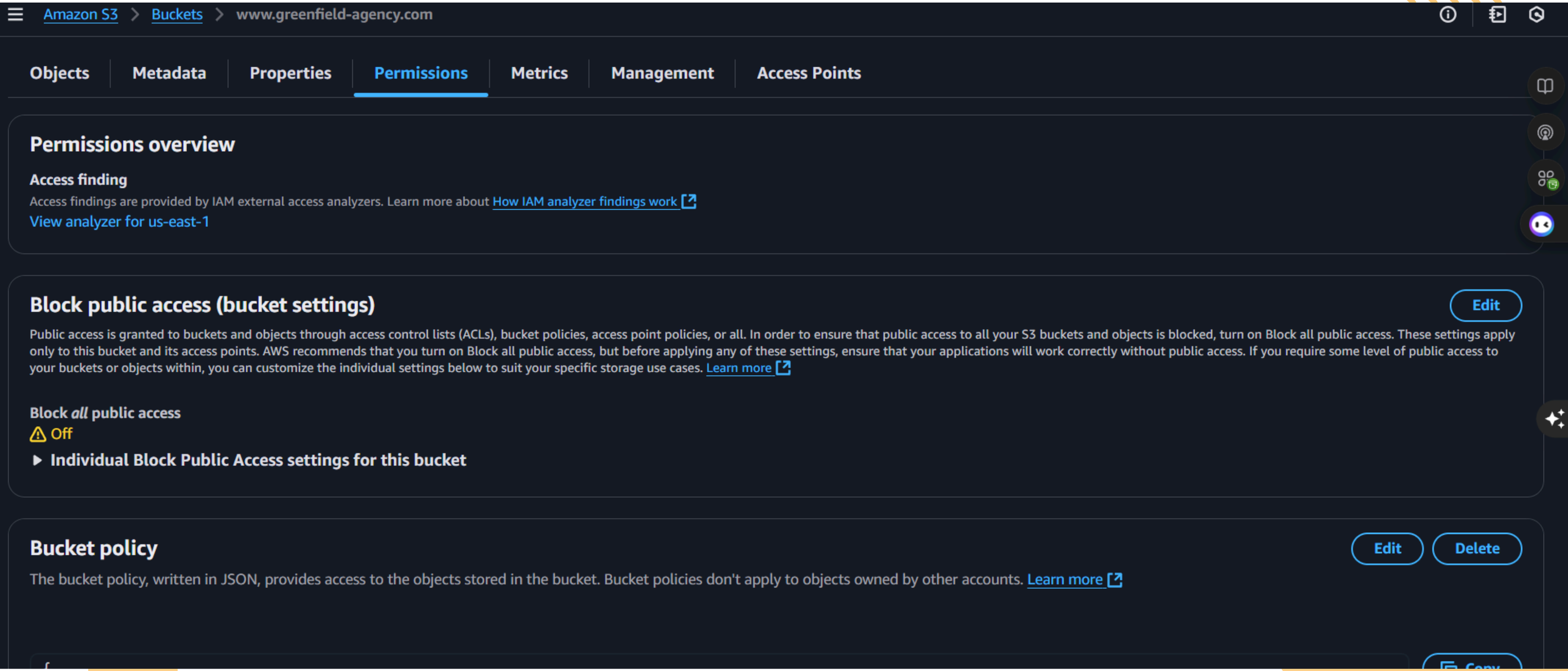

editing the properties of the s3 buckets




enabling of the static website




enabling and configuring the static website hosting




S3 bucket permission policy edit

## Bucket policy

Edit Delete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more

Copy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::www.greenfield-agency.com/*"
    }
  ]
}
```

S3 bucket permission policy edit

## Static website hosting

Edit

Use this bucket to host a website or redirect requests. Learn more

**We recommend using AWS Amplify Hosting for static website hosting**
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about Amplify Hosting or View your existing Amplify apps

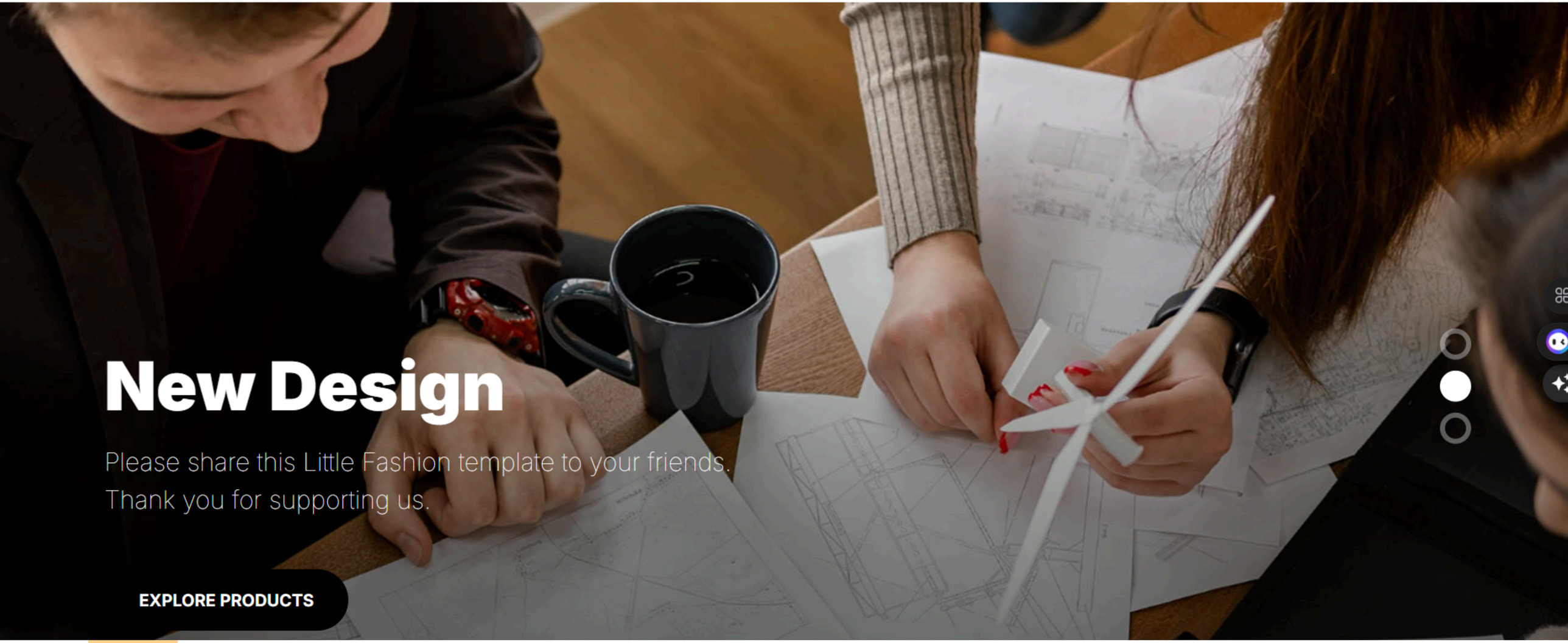
Create Amplify app

**S3 static website hosting**
Enabled

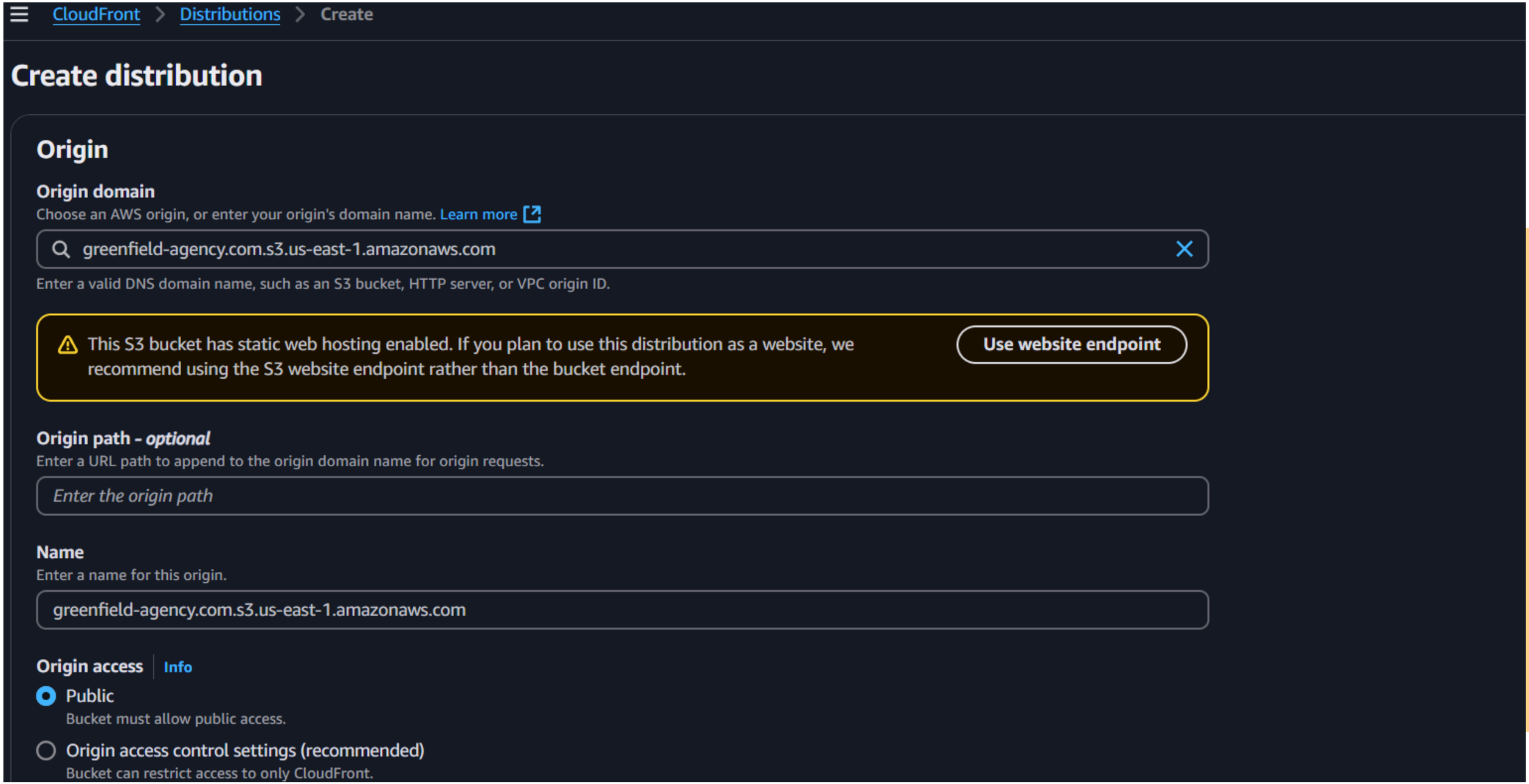
**Hosting type**
Bucket hosting

**Bucket website endpoint**
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more
http://www.greenfield-agency.com.s3-website-us-east-1.amazonaws.com

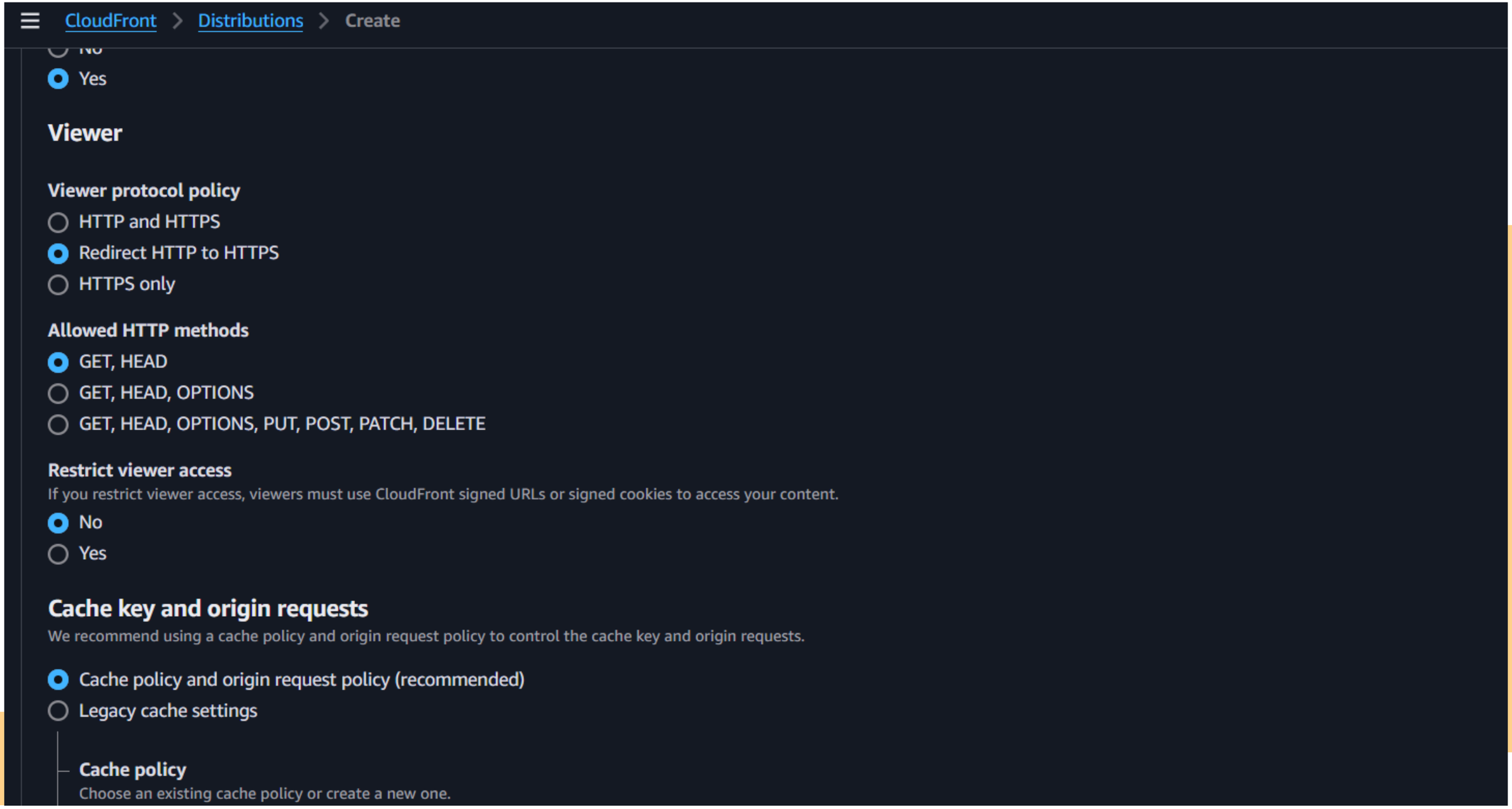
**static website link**



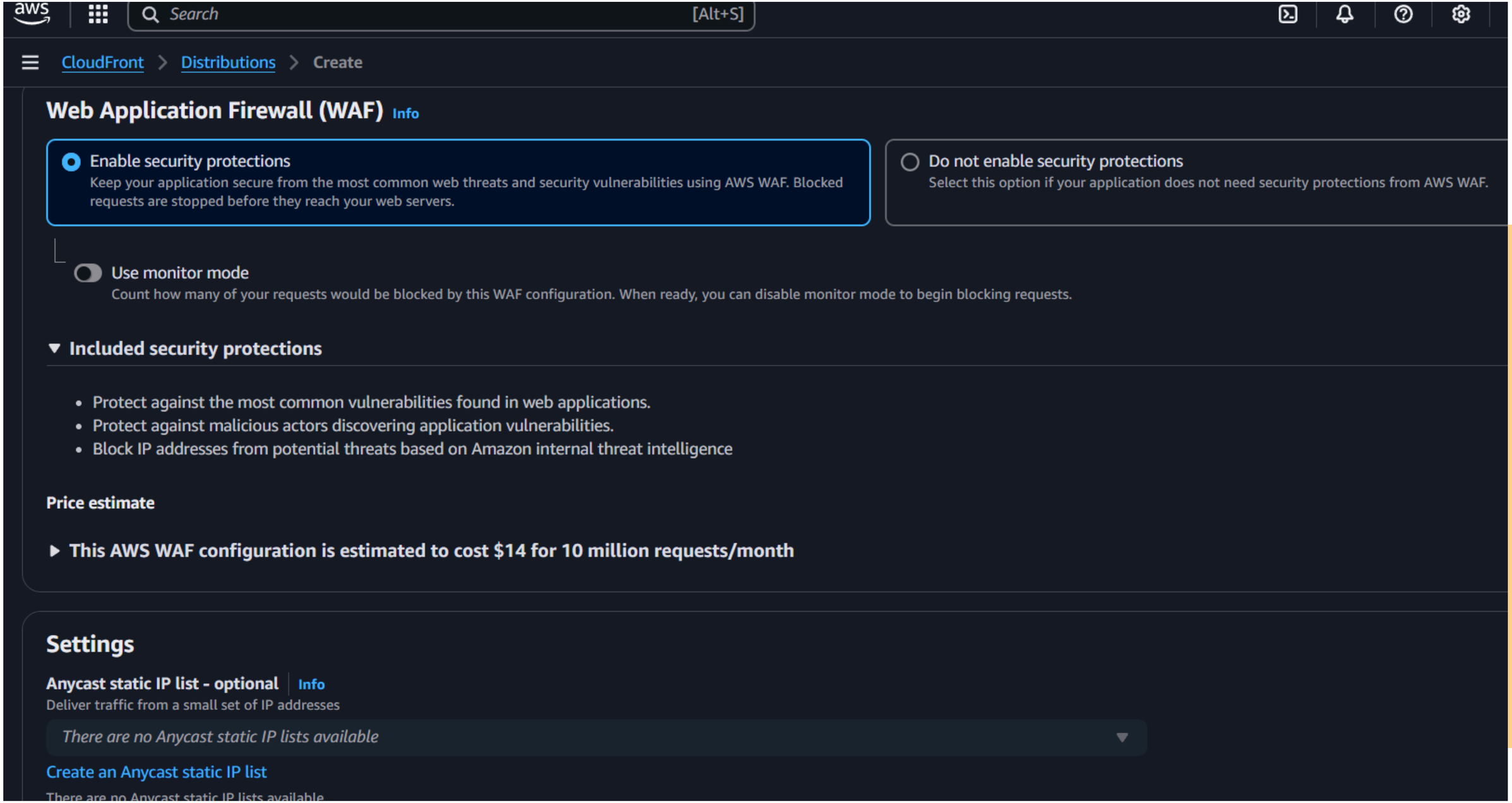

static website



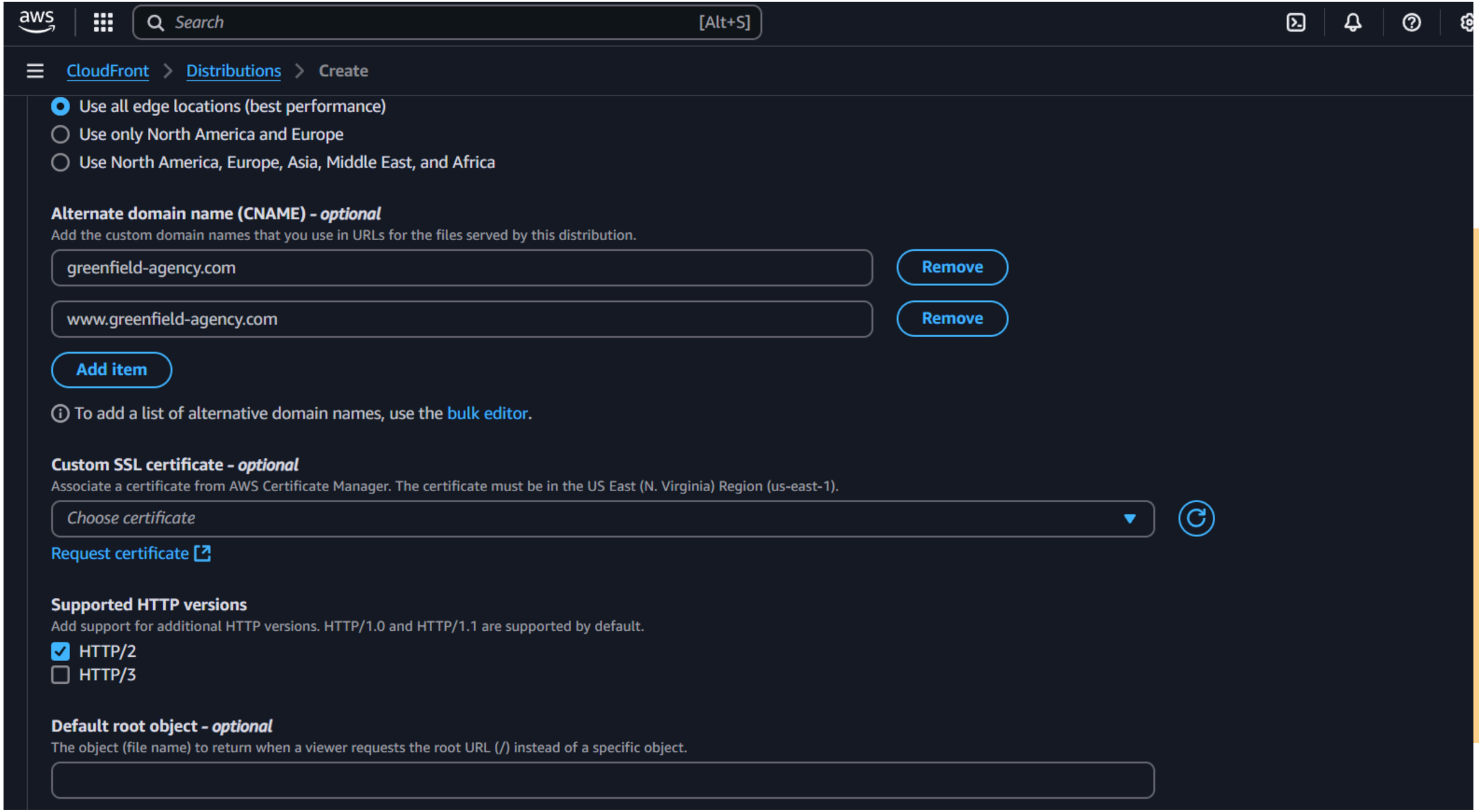

creating distribution



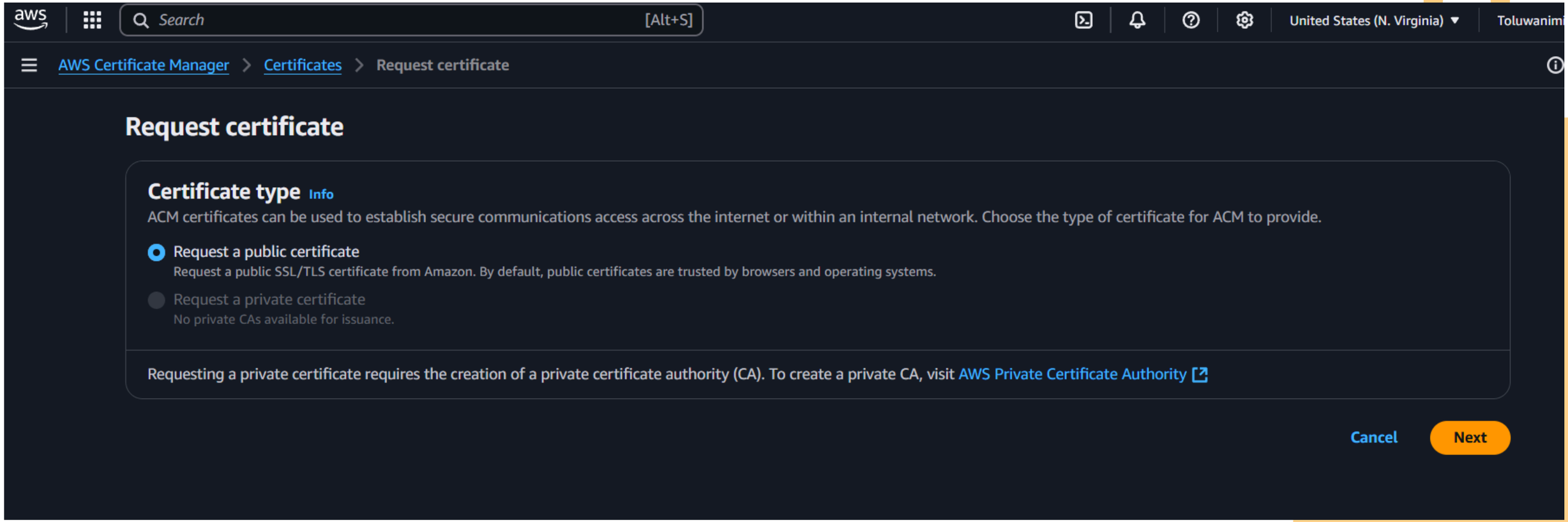

creating distribution



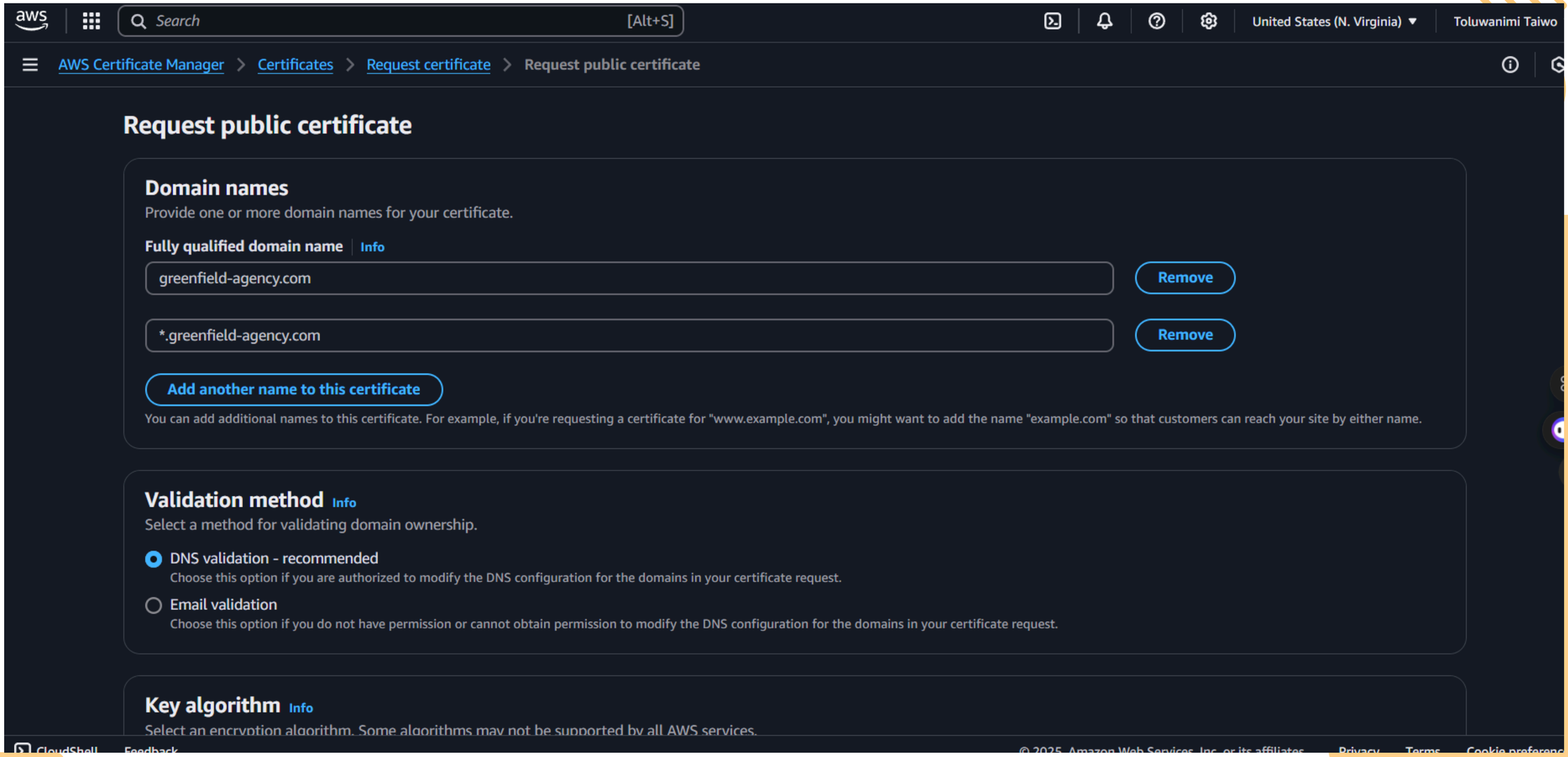

creating distribution



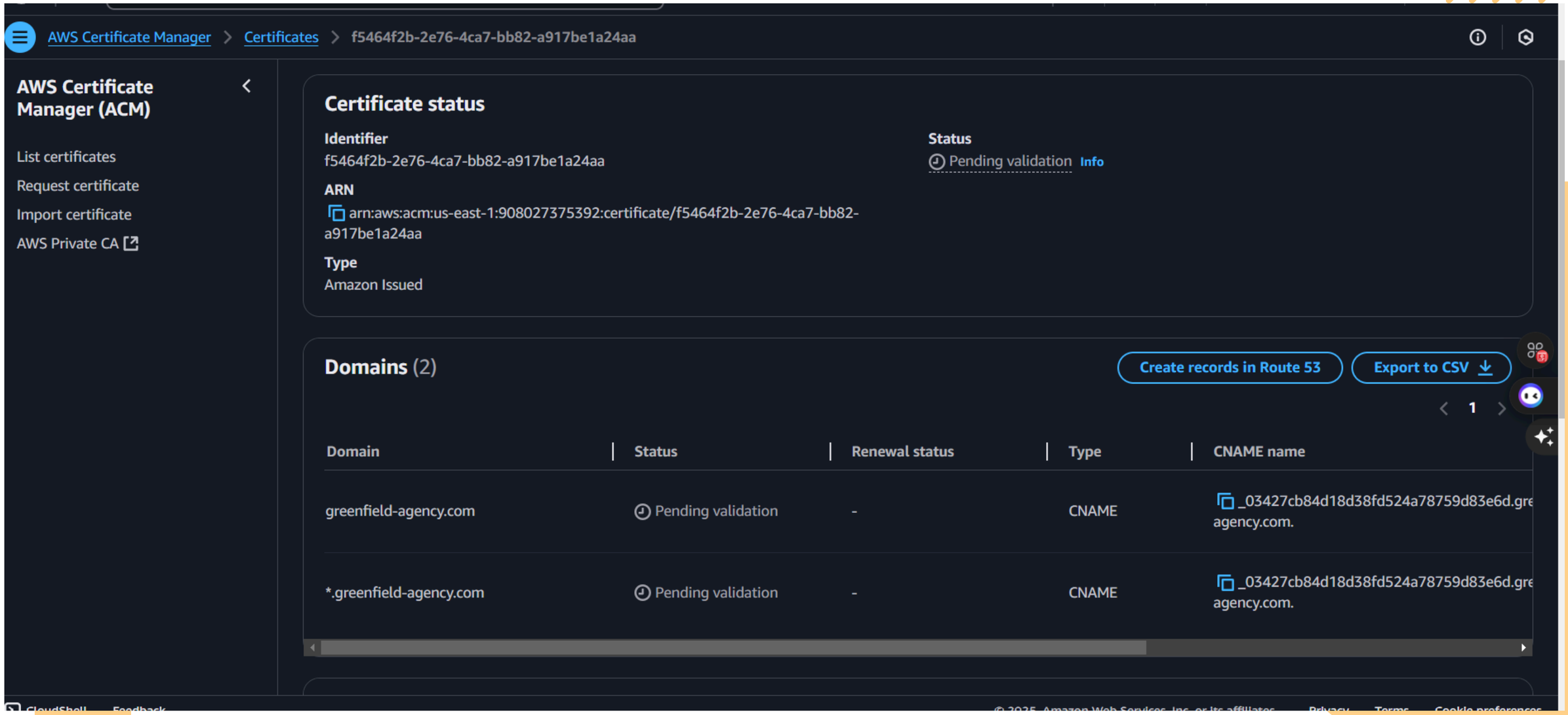

creating distribution



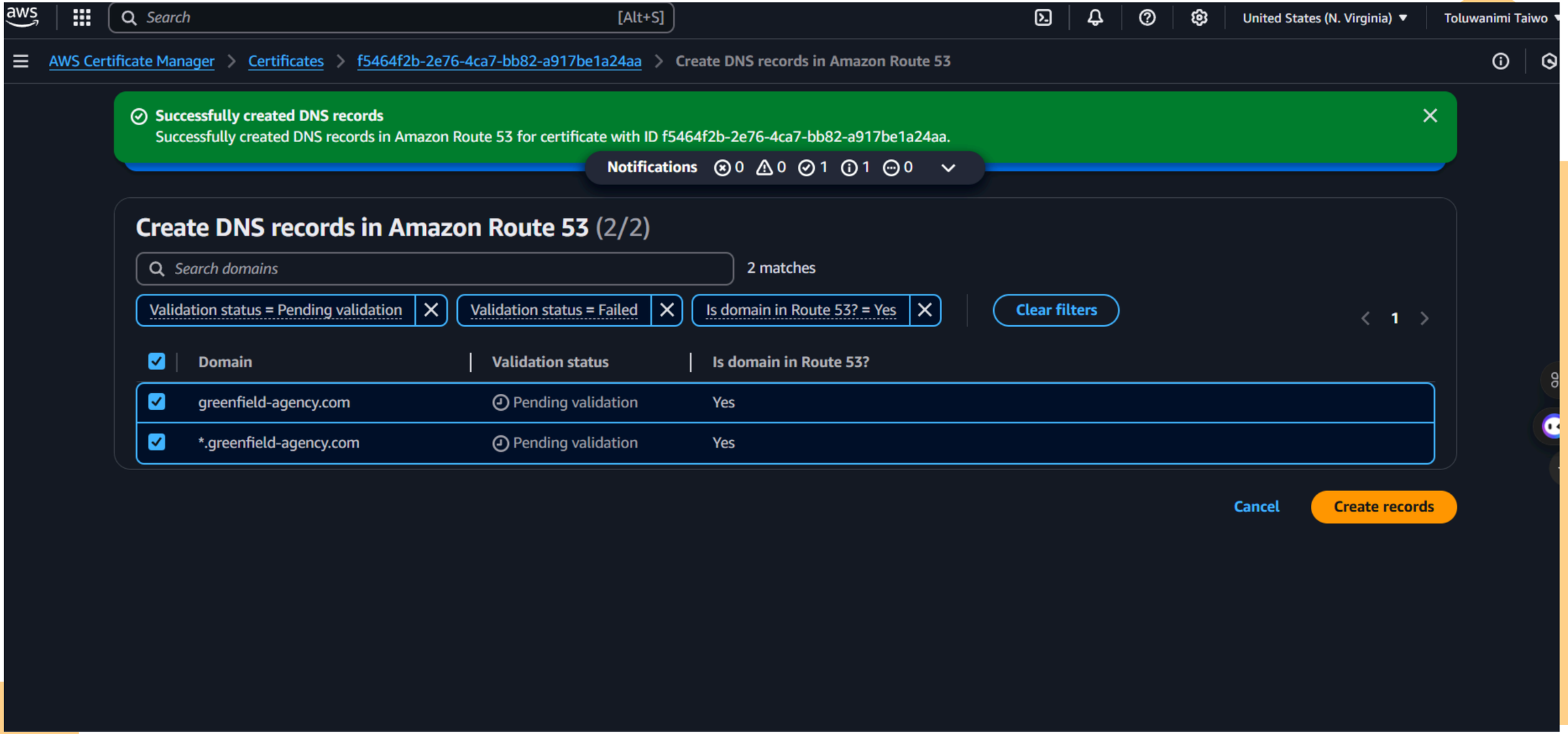

requesting for SSL certificate



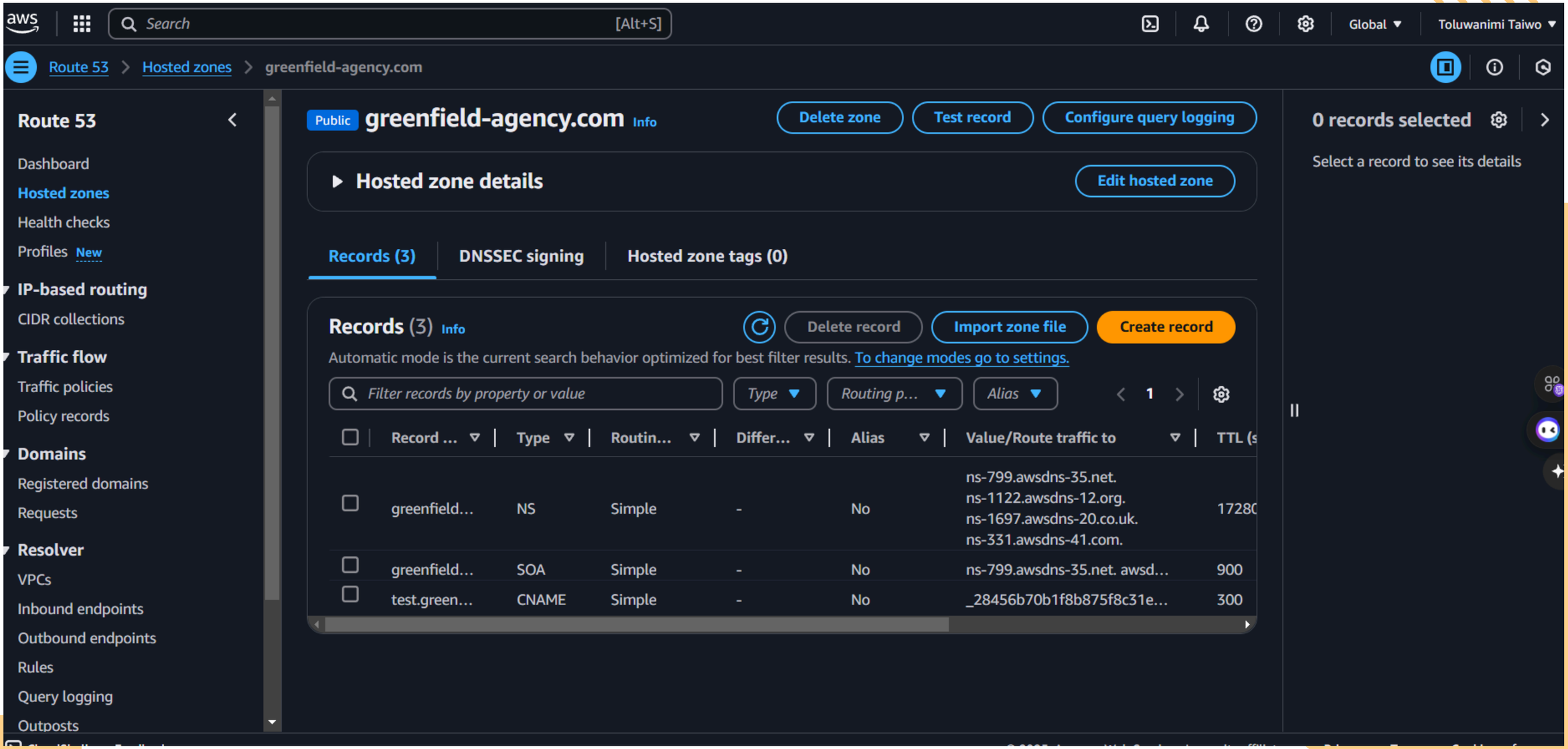

requesting for SSL certificate


AWS Certificate Manager > Certificates > f5464f2b-2e76-4ca7-bb82-a917be1a24aa
AWS Certificate Manager (ACM)
List certificates
Request certificate
Import certificate
AWS Private CA
Certificate status
Identifier
f5464f2b-2e76-4ca7-bb82-a917be1a24aa
Status
Pending validation Info
ARN
arn:aws:acm:us-east-1:908027375392:certificate/f5464f2b-2e76-4ca7-bb82-a917be1a24aa
Type
Amazon Issued
Domains (2)
Create records in Route 53
Export to CSV
1
Domain
Status
Renewal status
Type
CNAME name
greenfield-agency.com
Pending validation
-
CNAME
_03427cb84d18d38fd524a78759d83e6d.gre
agency.com.
*.greenfield-agency.com
Pending validation
-
CNAME
_03427cb84d18d38fd524a78759d83e6d.gre
agency.com.

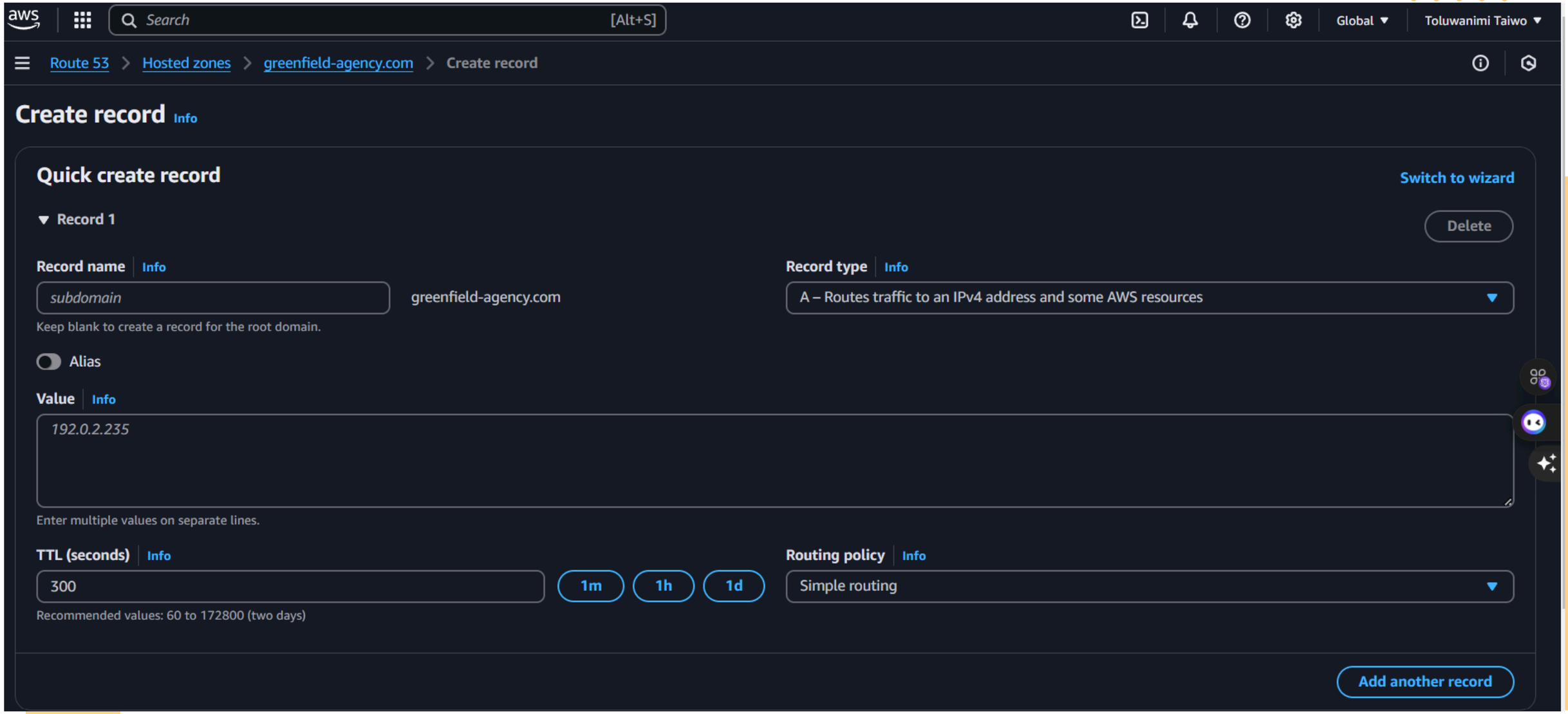
ACM


aws
Search
[Alt+S]
United States (N. Virginia)
Toluwanimi Taiwo
AWS Certificate Manager
Certificates
f5464f2b-2e76-4ca7-bb82-a917be1a24aa
Create DNS records in Amazon Route 53
Successfully created DNS records
Successfully created DNS records in Amazon Route 53 for certificate with ID f5464f2b-2e76-4ca7-bb82-a917be1a24aa.
Notifications 0 0 1 1 0
Create DNS records in Amazon Route 53 (2/2)
Search domains
2 matches
Validation status = Pending validation
Validation status = Failed
Is domain in Route 53? = Yes
Clear filters
1
Domain
Validation status
Is domain in Route 53?
greenfield-agency.com
Pending validation
Yes
*.greenfield-agency.com
Pending validation
Yes
Cancel
Create records

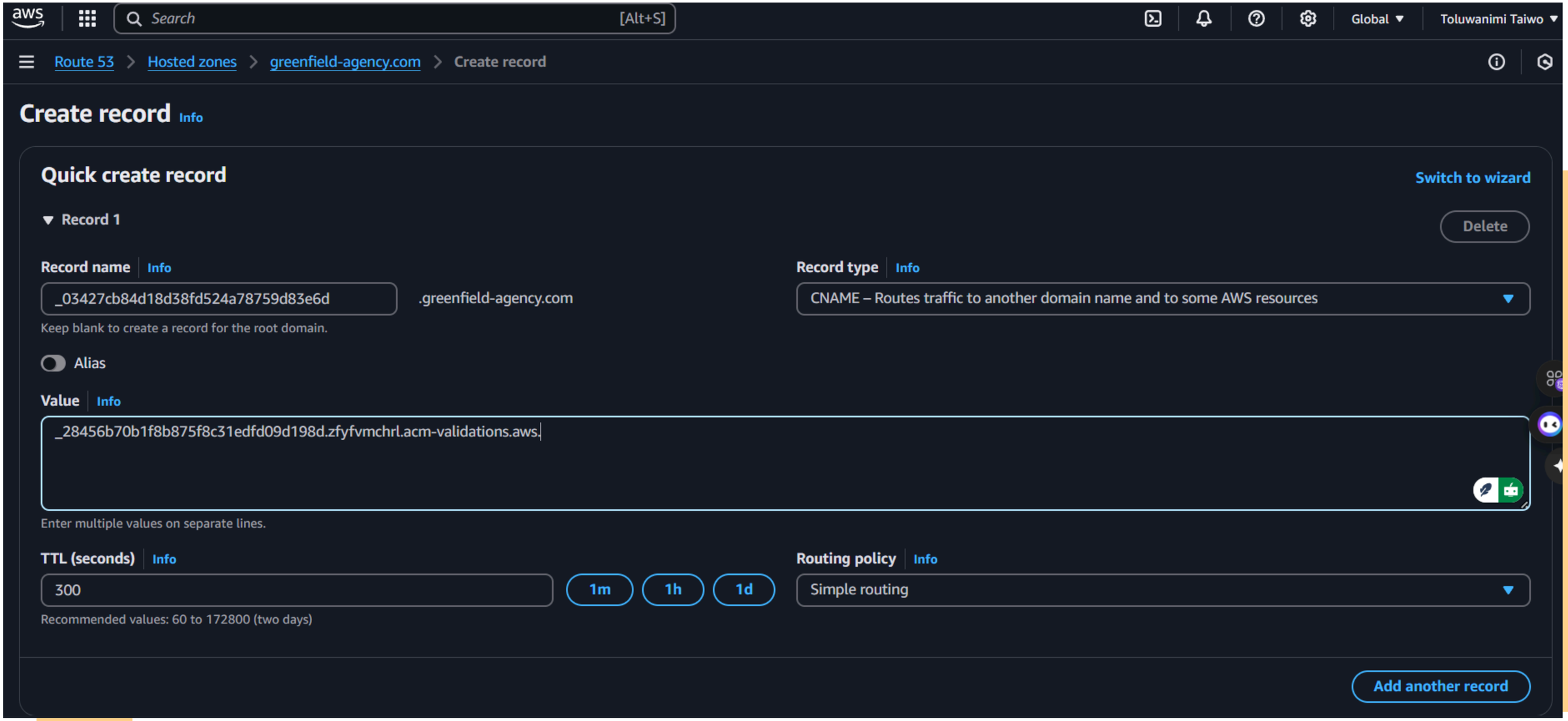
ACM


aws
Search
[Alt+S]
Global
Toluwanimi Taiwo
Route 53
Hosted zones
greenfield-agency.com
Route 53
Dashboard
Hosted zones
Health checks
Profiles New
IP-based routing
CIDR collections
Traffic flow
Traffic policies
Policy records
Domains
Registered domains
Requests
Resolver
VPCs
Inbound endpoints
Outbound endpoints
Rules
Query logging
Outposts
Public
greenfield-agency.com
Info
Delete zone
Test record
Configure query logging
Hosted zone details
Edit hosted zone
Records (3)
DNSSEC signing
Hosted zone tags (0)
Records (3)
Info
Delete record
Import zone file
Create record
Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.
Filter records by property or value
Type
Routing p...
Alias
1
Record ...
Type
Routin...
Differ...
Alias
Value/Route traffic to
TTL (s
greenfield...
NS
Simple
-
No
ns-799.awsdns-35.net.
ns-1122.awsdns-12.org.
ns-1697.awsdns-20.co.uk.
ns-331.awsdns-41.com.
1728(
greenfield...
SOA
Simple
-
No
ns-799.awsdns-35.net. awsd...
900
test.green...
CNAME
Simple
-
No
_28456b70b1f8b875f8c31e...
300
0 records selected
Select a record to see its details

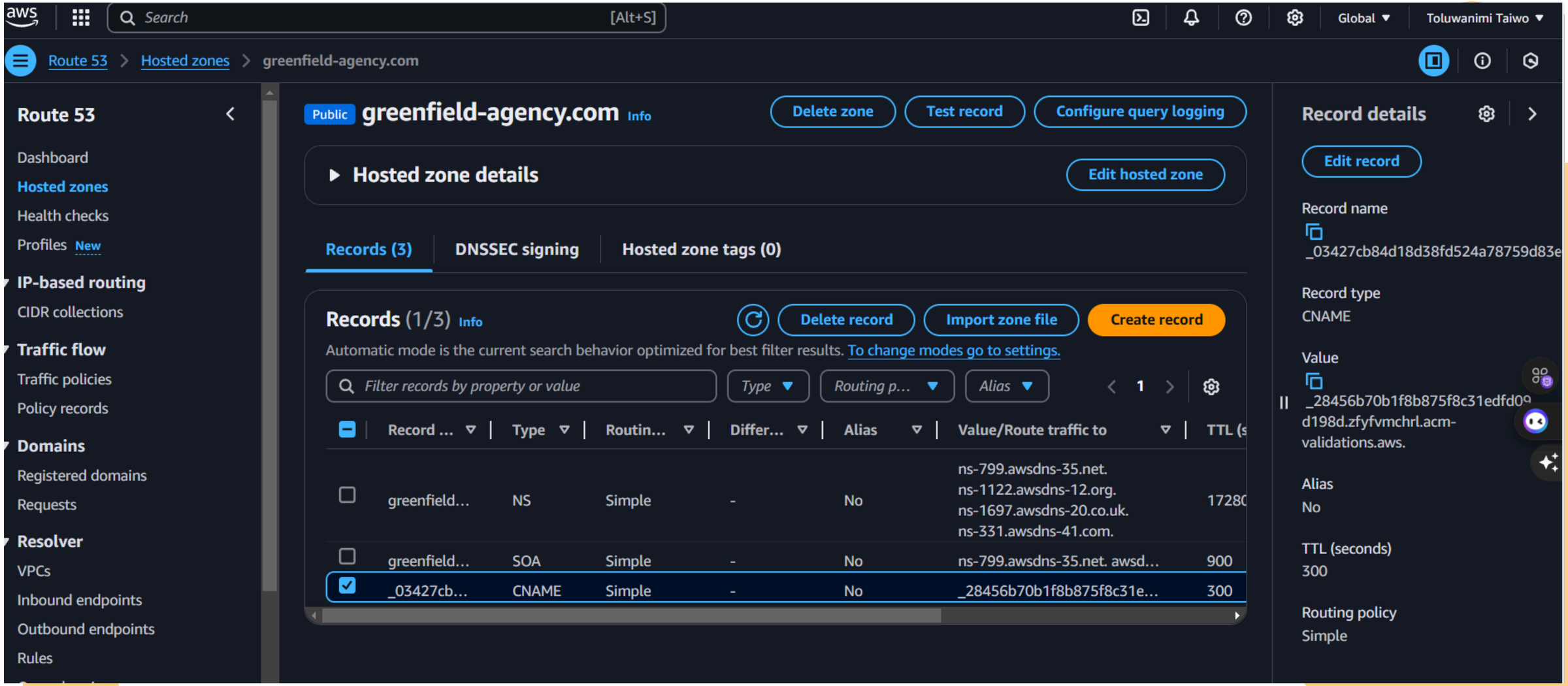
ROUTE53


aws
Search
[Alt+S]
Global
Toluwanimi Taiwo
Route 53
Hosted zones
greenfield-agency.com
Create record
Create record Info
Quick create record
Switch to wizard
Record 1
Delete
Record name Info
subdomain
greenfield-agency.com
Keep blank to create a record for the root domain.
Record type Info
A – Routes traffic to an IPv4 address and some AWS resources
Alias
Value Info
192.0.2.235
Enter multiple values on separate lines.
TTL (seconds) Info
300
1m
1h
1d
Recommended values: 60 to 172800 (two days)
Routing policy Info
Simple routing
Add another record

ROUTE53


aws
Search
[Alt+S]
Global
Toluwanimi Taiwo
Route 53
Hosted zones
greenfield-agency.com
Create record
Create record Info
Quick create record
Switch to wizard
Record 1
Delete
Record name Info
_03427cb84d18d38fd524a78759d83e6d
.greenfield-agency.com
Keep blank to create a record for the root domain.
Record type Info
CNAME – Routes traffic to another domain name and to some AWS resources
Alias
Value Info
_28456b70b1f8b875f8c31edfd09d198d.zfyfvmchrl.acm-validations.aws.
Enter multiple values on separate lines.
TTL (seconds) Info
300
1m
1h
1d
Recommended values: 60 to 172800 (two days)
Routing policy Info
Simple routing
Add another record

ROUTE53


aws
Search
[Alt+S]
Global
Toluwanimi Taiwo
Route 53
Hosted zones
greenfield-agency.com
Route 53
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Public greenfield-agency.com Info
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1
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Type
Routin...
Differ...
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Value/Route traffic to
TTL (s
greenfield...
NS
Simple
-
No
ns-799.awsdns-35.net.
ns-1122.awsdns-12.org.
ns-1697.awsdns-20.co.uk.
ns-331.awsdns-41.com.
17280
greenfield...
SOA
Simple
-
No
ns-799.awsdns-35.net. awsd...
900
_03427cb...
CNAME
Simple
-
No
_28456b70b1f8b875f8c31e...
300
Record details
Edit record
Record name
_03427cb84d18d38fd524a78759d83e
Record type
CNAME
Value
_28456b70b1f8b875f8c31edfd09
d198d.zfyfvmchrl.acm-
validations.aws.
Alias
No
TTL (seconds)
300
Routing policy
Simple

ROUTE53